



INSTALLATION INSTRUCTIONS



AGC-4



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5. Technical specifications

1. General information

1.1 Warnings, legal information and safety

1.1.1 Warnings and notes

Throughout this document, a number of warnings and notes with helpful user information will be presented. To ensure that these are noticed, they will be highlighted as follows in order to separate them from the general text.

Warnings



DANGER!

This highlights dangerous situations. If the guidelines are not followed, these situations could result in death, serious personal injury, and equipment damage or destruction.



CAUTION

This highlights potentially dangerous situations. If the guidelines are not followed, these situations could result in personal injury or damaged equipment.

Notes



INFO

Notes provide general information, which will be helpful for the reader to bear in mind.

1.1.2 Legal information and disclaimer

DEIF takes no responsibility for installation or operation of the generator set or switchgear. If there is any doubt about how to install or operate the engine/generator or switchgear controlled by the Multi-line 2 unit, the company responsible for the installation or the operation of the equipment must be contacted.



INFO

The Multi-line 2 unit is not to be opened by unauthorised personnel. If opened anyway, the warranty will be lost.

Disclaimer

DEIF A/S reserves the right to change any of the contents of this document without prior notice.

The English version of this document always contains the most recent and up-to-date information about the product. DEIF does not take responsibility for the accuracy of translations, and translations might not be updated at the same time as the English document. If there is a discrepancy, the English version prevails.

1.1.3 Safety issues

Installing and operating the Multi-line 2 unit may imply work with dangerous currents and voltages. Therefore, the installation should only be carried out by authorised personnel who understand the risks involved in working with live electrical equipment.



DANGER!

Be aware of the hazardous live currents and voltages. Do not touch any AC measurement inputs as this could lead to injury or death.

1.1.4 Current transformer danger



DANGER!



Electrical shock and arc flash

Risk of burns and electrical shock from high voltage.

Short all current transformer secondaries before breaking any current transformer connections to the controller.

1.1.5 Electrostatic discharge awareness

Sufficient care must be taken to protect the terminal against static discharges during the installation. Once the unit is installed and connected, these precautions are no longer necessary.

1.1.6 Factory settings

The controller is delivered pre-programmed from the factory with a set of default settings. These settings are based on typical values and may not be correct for your system. You must therefore check all parameters before using the controller.

1.2 About the installation instructions

1.2.1 General purpose

These Installation Instructions include hardware information, mounting instructions, terminal strip descriptions, I/O lists and wiring descriptions.

The purpose of this document is to give the user important information to be used in the installation of the controller.



CAUTION

Read this document before starting to work with the Multi-line 2 unit and the genset to be controlled. Failure to do this could result in human injury or damage to the equipment.

1.2.2 Intended users

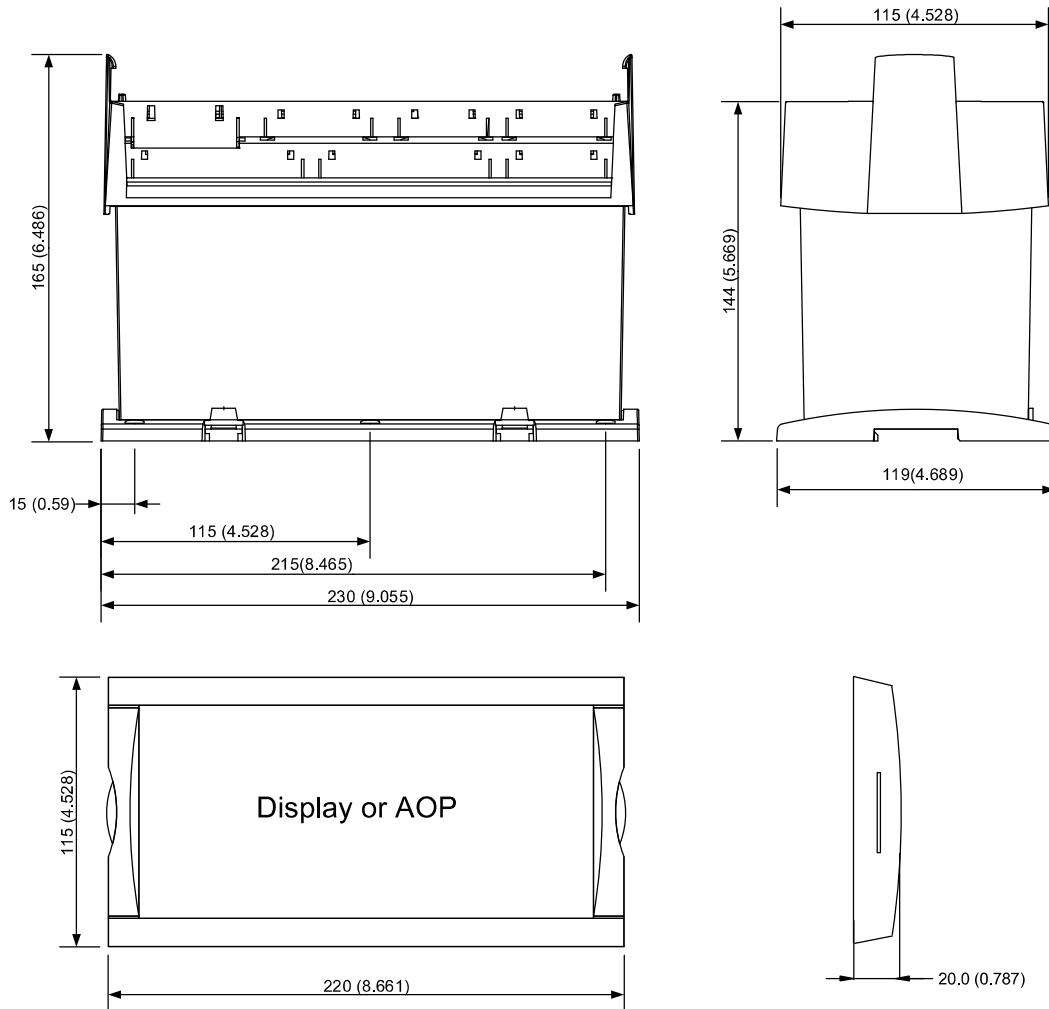
These Installation Instructions are mainly intended for the person responsible for the design and installation. In most cases, this would be a panel builder designer. Naturally, other users might also find useful information in the document.

1.2.3 Drawings

Refer to www.deif.com for the most recent 3D-files, drawings, E-drawings and E-plans.

2. Mounting

2.1 Dimensions

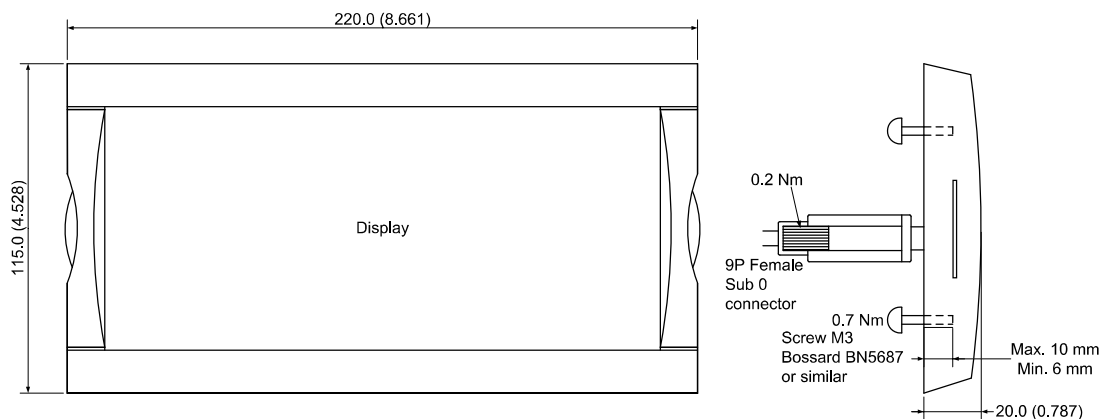


INFO

Dimensions are given in mm (inches).

2.2 Tightening torques

Controller:	1.5 Nm for the six M4 screws (countersunk screws are not to be used)
Plug connections (terminals):	0.5 Nm, 4.4 lb-in
DU-2/AOP-1/AOP-2 (see diagram below)	
Panel door mounting:	0.7 Nm, 6.2 lb-in
D-sub screw:	0.2 Nm, 1.8 lb-in
DC-DC converter terminals:	0.5 Nm, 4.4 lb-in



2.3 Mounting of the equipment

The controller is designed for mounting inside the panel. The DU-2 display can be installed on the panel door and connected to the controller with a display cable.

2.3.1 Controller mounting

The controller can be mounted:

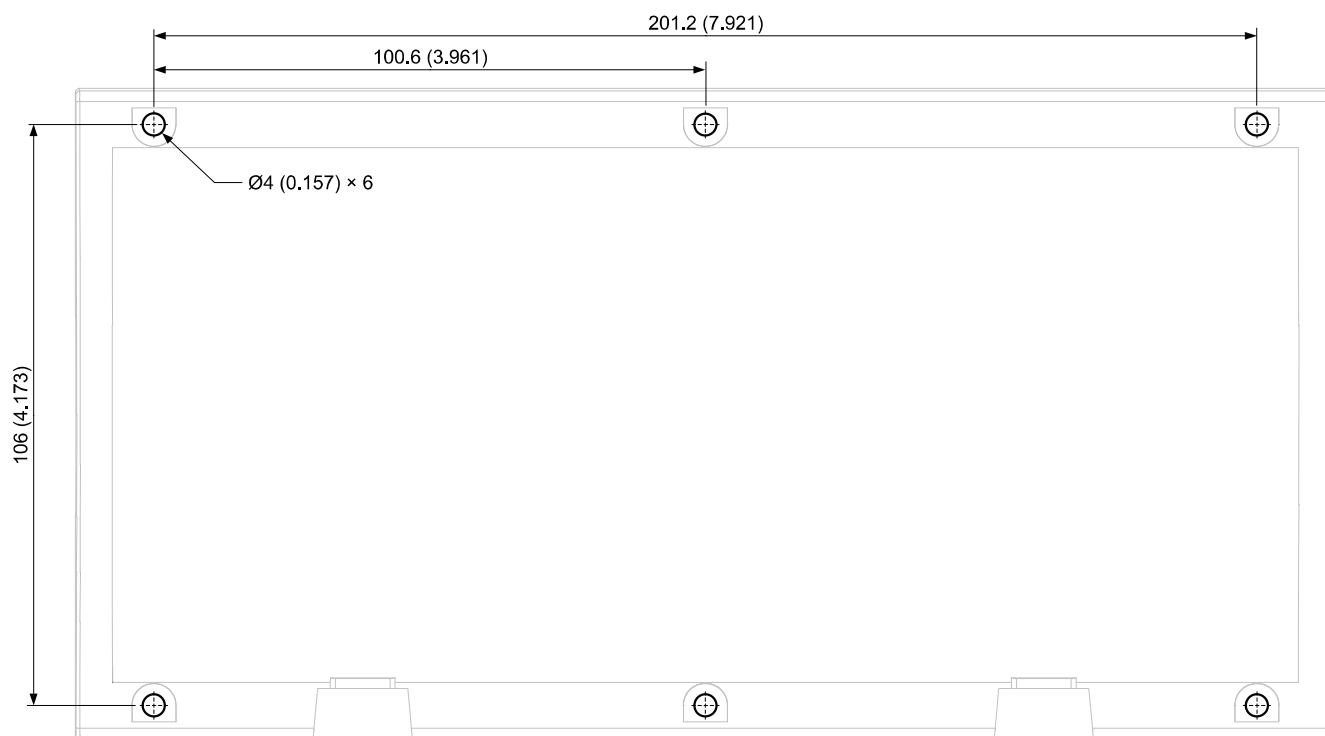
1. With screws to the rear side of the cabinet. Six screw holes are available for this.
2. Directly on a DIN rail.



INFO

DEIF recommends using the screw hole fastening.

2.3.2 Controller mounting drilling diagram

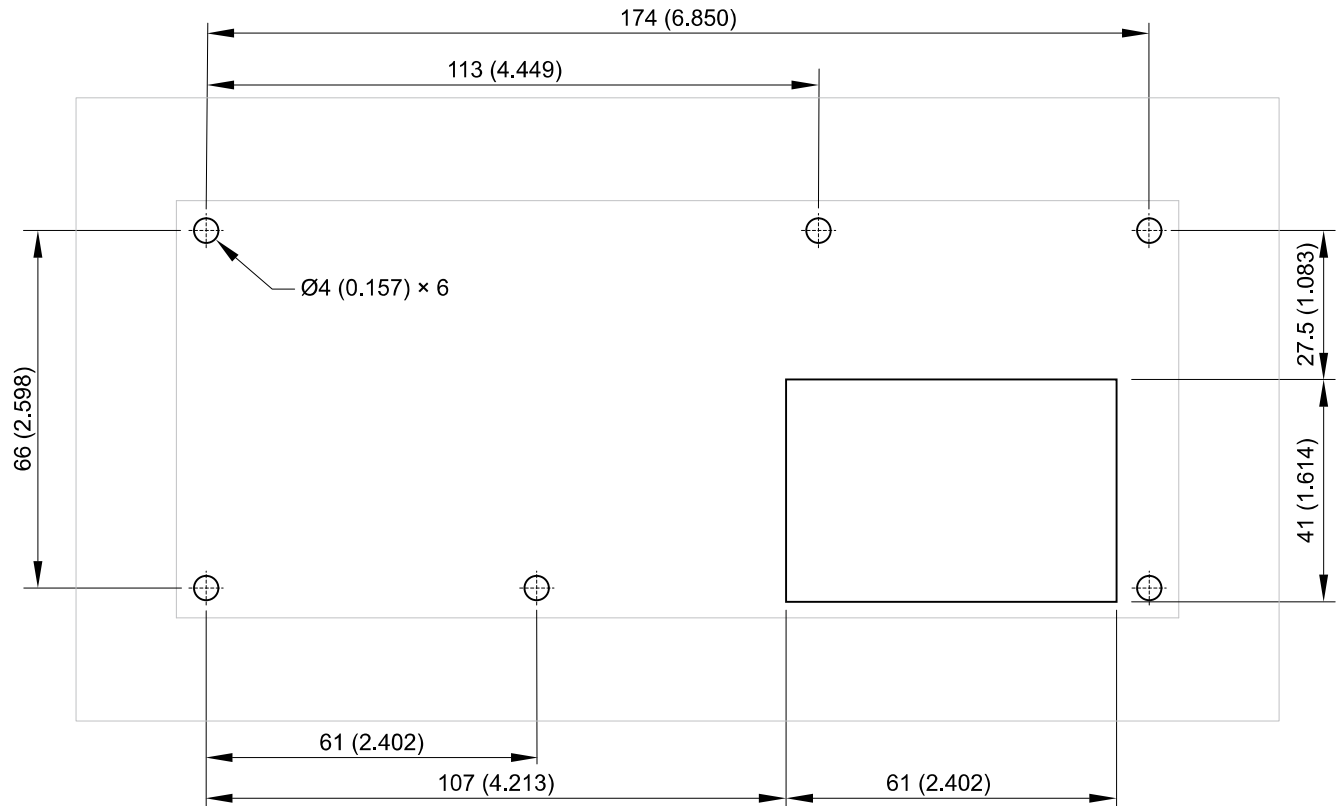


**INFO**

Measurements are in mm (inches).

2.3.3 DU-2/AOP display panel cutout

Cut and drill the panel door for the DU-2/AOP according to the diagram below.

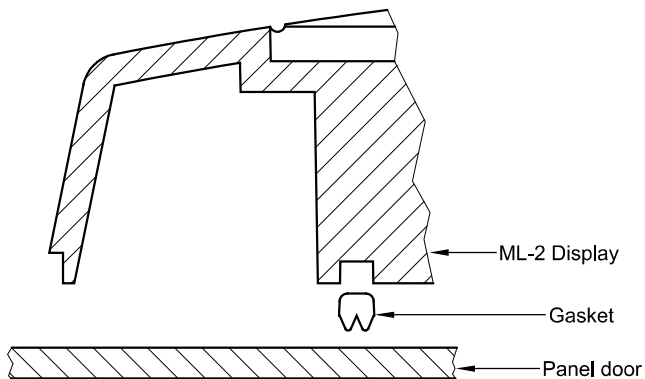
**INFO**

Dimensions are in mm (inches).

2.3.4 Mounting of DU-2/AOP gasket (option L1)

It is important to mount the gasket correctly to have an IP65 seal. You must use all six DU-2 or AOP screws to ensure IP65.

Mount the gasket as shown below.



3. Hardware

3.1 Board slot positions

The unit housing is divided into board slot positions. This means that the unit consists of a number of printed circuit boards (PCB) mounted in numbered slots. The green terminal blocks are then mounted in the PCBs. Some of these board slots are standard and some are intended for options. The board slot positions are arranged as illustrated below.

Slot type	Option	Slot #1	Slot #3	Slot #5	Slot #7
Terminals		1-28	37-64	73-89	98-125
Power supply	Standard	X			
AC measurements	Standard			X	
Engine interface	Standard/M4				X
Load sharing	G3		X		
Power management	G4/G5/G8				X
Engine communication	H7				X
I/O extension	M12		X		

Slot type	Option	Slot #2	Slot #4	Slot #6	Slot #8
Terminals		29-34	65-72	90-97	126-133
Analogue controller outputs	E1/E2		X		
Analogue transducer outputs	F1			X	
Combination outputs	EF2/EF4/EF5/EF6		X		
Serial communication	H2/H3/H9	X			
Engine communication	H5/H6/H13				X
Engine communication and external I/O	H12.2/H12.8	X			X
I/O extension cards	H8.2/M13.2/M14.2	X			
I/O extension cards	M13.6/M14.6/M15.6/M16.6			X	
I/O extension cards	H8.8/M13.8/M14.8/M15.8/M16.8				X



INFO

Only hardware options, which will affect the hardware of the unit, are represented in the table. The software options will be seen through the PC utility software. The software options that are not represented in the above table can be found in the data sheet.